

Grid Stress under Power Saver Plan

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1 Introduction

We want to show the benefits of allowing retailers and distribution network service providers (DNSPs) to choose which 3-hour period during the day a customer gets free electricity.

The goal is to provide economic fairness to customers without solar panels within network constraints, rather than to maximize network efficiency. However, network efficiency cannot be ignored. The tariff must be very simple.

2 Trivial case

Assume that most customers have moderate demands, but that some small fraction p have very large demands during the peak time. Call these customers type L customers. In particular, assume that a distribution network can support one type L customer at a time, but not two.

In a network with N customers, the probability of having i type L customers is given by the binomial formula

$$p_i = \binom{N}{i} p^i (1-p)^{N-i} \quad (1)$$

If all customers have the same free time, the probability that the network is overloaded is

$$p_2 = \binom{N}{2} p^2 (1-p)^{N-2} \quad (2)$$

whereas if the network can choose one of two non-overlapping free times, say either 9:30am–12:30pm or 12:30pm–3:30pm, then it can support up to two type L customers, by allocating one to each free slot. The probability that the network is overloaded is then the probability of three type- L customers,

$$p_3 = \binom{N}{3} p^3 (1-p)^{N-3}. \quad (3)$$

Specifically,

$$\frac{p_2}{p_3} = \frac{3}{(N-3)} \frac{1-p}{p} \quad (4)$$

This can be quite a big reduction. For example, if $N = 33$ and $p = 10^{-3}$, $p_2/p_3 = 100$. (This example has p too small, so I could do the arithmetic in my head. We should probably choose numbers so the benefit is about a factor of 10 and the probability of a problem is at least 0.01. Perhaps $N = 30$ and $p = 0.003$ may work.) That means that 100 times fewer networks would require upgrades, simply by allowing the retailer and network provider a choice of which times in the middle of the day they offer free.

3 More realistic case

Consider a collection of networks, with given line impedances (edge weights). Randomly place a Poisson number of type L customers in the network and see how often the voltage drop becomes a problem. It is a problem if the voltage drop is bigger than, say, 20 V. I'll have to talk to someone else to find suitable parameters for line impedance, but I think it is of the order of 0.1Ω per line segment. To work out the voltage drop, assume that each customer has 240 V, and power about 24 kW, so current 100 A. The voltage drop is then the sum over a path from the node to the root of the impedance of a line segment times the total current draw of all customers downstream of it (i.e., further from the root).

We can either do this analytically or by Monte Carlo simulation. To do it analytically, I'd start with a very simple tree, like just a stick (string of nodes with degree 2). We could then move to things like a balanced 4-leaf tree, except with the edges replaced by sticks with, say, 4 degree-two customers per stick (excluding the degree-one and degree-three customers).

4 Role of solar customers

Increasing demand is intended to offset solar generation on the same feeders. However, the problems of excess load occur on cloudy days, when solar generation is negligible. The DNSP must provide for safe operation even when there is no solar generation during the free window. (A much better engineering solution, although more complicated for customers, would be to guarantee free electricity on all but a small number of days per year which the DNSP can choose, allowing the demand to be curtailed during these heavily overcast days.)

5 Convenience to the customer

The morning/afternoon times may be more convenient for a customer, especially if the high load is for EV charging, since the requirement of being home from 11am-2pm makes it hard to do half-day tasks in both the morning and in the afternoon, whereas one of these is possible if the free time is shifted.

In some cases, customers may also be given the choice of whether they want free time in the morning or afternoon. If customers get total choice, the benefits for the network are much less (essentially replacing N by $N/2$ in the p_2 formula, rather than using the p_3 formula), and so that is not desirable. If some customers get a choice and others don't, that would lead to dissatisfaction. For this reason, a retailer would have to consider carefully how to market any flexibility given to the customers.

6 Effect on generation, transmission and MV distribution

It is unclear how many customers would take up the free electricity.

If the uptake is so large that it causes peak demand to occur during the free time, then it is advantageous to spread the free time over a longer interval.

If the uptake is small enough that peak demand remains in the morning and evening, then having the free interval encroach closer on those peaks may increase the peak demand for generation and transmission.

If the retailers have a choice between three intervals, with two not overlapping and one in the middle of the day overlapping both, then the shape of the new demand can be controlled by choosing how many customers to allocate to each interval.